

High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks

Minchan Kim, Kiwan Kim and Jiyun Lee

Korea Advanced Institute of Science and Technology

Sam Pullen

Stanford University

Recent technological improvements and increased military utilization have proven the operational viability of unmanned aerial vehicles (UAVs) and made them attractive for a wide range of potential civil applications. In particular, the capability and low cost of small UAVs have led to a dramatic growth in commercial applications over the past few years as well as generating major interest in the media. While most current UAVs are remotely piloted by humans, future UAVs will operate autonomously and coordinate their activities in large groups. Autonomous UAVs networks guided by central controllers will safely and routinely perform various missions such as surveillance, reconnaissance, and data collection [1].

Prior work [1, 2] has proposed a concept of UAV network operations that use differential corrections from Local-Area Differential GNSS (LADGNSS) to provide increased accuracy, safety, and reliable navigation to UAVs within 5 to 100 km of the controller station. LADGNSS supports low cost for commercial applications and high integrity to allow UAVs to operate in close proximity to each other while minimizing collision risk. Local-area UAV networks can perform their tasks safely as long as UAV locations within the network are monitored by the controller to insure that safe separation distances are maintained from other vehicles and ground obstructions. The controller, including a LADGNSS reference station, receives GNSS signals and positions of UAVs, estimates errors and separation distances, and broadcasts differential corrections and optimized guidance for precise and safe operation.

This paper presents a methodology to estimate the minimum separation distance that should be defined to prevent collisions as part of autonomous UAV guidance. Separation distance can be derived by modeling

mainly two sources of position uncertainty that could potentially lead to collisions: (i) navigation error, or the residual error in the estimated position (compared with the true position), which cannot be completely eliminated by LADGNSS; and (ii) path-following error, or error in following the path defined by the guidance source, which depends on the controller's performance and external disturbances such as wind. This paper estimates these errors using standard LADGNSS error models developed for manned aircraft and empirical bounds derived from data collected from UAV flight tests.

Residual navigation error after applying the differential corrections provided by LADGNSS consists of four error terms: residual tropospheric error, residual ionospheric error, ground facility error, and airborne error, which is the RSS (Root Sum Square) of airborne multipath error and receiver noise. Ground facility errors included in the position information of UAVs within the same network are very highly correlated when UAVs apply differential corrections from the same ground station at the same time. For these reasons, when we estimate separation distances between UAVs within the same network, ground facility errors are assumed to cancel out. Residual tropospheric and ionospheric errors are small and can be calculated by theoretical models given approximate UAV positions and velocities. Meanwhile, UAV multipath error should be analyzed with empirical data, since this error is more strongly affected by the flight environment than that of large manned aircraft. We conducted flight experiments with an octocopter (measuring 130 cm in diameter, 45 cm in height, and 5.26 kg in weight) equipped with a GNSS receiver and antenna to collect GNSS measurements and then estimated the multipath errors of the UAV. The results show that, under the flight conditions tested, the standard multipath error model for manned aircraft also bounds the UAV multipath error with margin.

Path-following error, which depends on the controller mounted on the UAV, was also analyzed based on position data collected during these flight tests. Path-following error is conservatively estimated by subtracting the LADGNSS position solution obtained from the UAV controller from the ideal position selected by the controller. Note that LADGNSS position is used in place of the (unknown) true position that normally defines path-following

error. This means that the path-following error estimate unavoidably includes a component of navigation sensor error as well.

Through an analysis of the path-following error for straight flight path cases, we determined that a lateral error bound of 2.3 m slightly exceeds the largest path-following error after adding a margin of about 19% to the largest error that occurred. The proposed error models were used to calculate specific minimum separation distances that satisfy the probabilities obtained from a risk-assessment methodology (derived from [2, 3, 4, 5]) to ensure the safety of UAV operations. Controllers that plan the path of UAVs within the same network based on the resulting separation distances also help to protect non-participating aircraft and humans from potential collisions.

An LADGNSS architecture based upon a simplification of the Ground Based Augmentation System (GBAS) has been proposed to provide high-integrity differential corrections to UAV networks with low cost and significantly lower complexity [1, 2]. This low-cost ground system will suffer higher GNSS errors and potentially limit the benefit of reducing UAV errors. If the limitations of a low-cost LADGNSS system are constraining, performance and integrity can be recovered by incorporating SBAS or an airborne-to-ground component of the two-way datalink. The paper describes modifications to LADGNSS in order to retain as much performance of GBAS as is feasible in the context of UAV navigation. This paper also tests the performance of LADGNSS after applying the proposed modifications in a UAV network.

References:

- [1] S. Pullen, P. Enge, and J. Lee, "High-Integrity Local-Area Differential GNSS Architectures Optimized to Support Unmanned Aerial Vehicles (UAVs)," *Proceedings of ION ITM 2013*, San Diego, CA, Jan. 28-30, 2013.
- [2] S. Pullen, "Managing Separation of Unmanned Aerial Vehicles Using High-Integrity GNSS Navigation," *Proc. EIWAC 2013*, Tokyo, Japan, Feb. 19-21, 2013.

- [3] *FAA System Safety Handbook*. Washington, D.C., U.S. Dept. of Transportation/Federal Aviation Administration, Dec. 2000, Ch. 3, "Principles of System Safety."
- [4] D. Gebre-Egziabher, Z. Xing, *Analysis of Unmanned Aerial Vehicles: Concept of Operations in ITS Applications*, Final Report (CTS 11-06), Center for Transportation Studies, University of Minnesota, March 2011.
- [5] C.W. Lum, B. Waggoner, "A Risk Based Paradigm and Model for Unmanned Aerial Systems in the National Airspace," *Proc. AIAA Infotech@Aerospace 2011 Conference*, St. Louis, MO, March 29-31, 2011, AIAA-2011-1424.

What is new and innovative?

This paper develops a new methodology to estimate the minimum separation distance between UAVs supported by LADGNSS within the same autonomous network. Position uncertainty is modelled by using empirical data collected from UAV flight tests and standard LADGNSS error models. The paper also describes modifications to LADGNSS in order to retain as much of the performance of GBAS as is feasible in the context of UAV navigation.

Session choices:

Aviation and Marine Applications

[GNSS Error Models](#)

[High Precision GNSS](#)